



Colour Identification Using TCS3200 Module

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This project is used to detect colours and display the colour names on an LCD using the TCS3200 colour sensor module and Arduino board. The sensor outputs red, green, and blue colours' reflected light values using colour tinted photo-diodes. The colour output comes as a frequency, and the value depends on the intensities.

Theory of colour identification

To understand what the eye perceives as a particular colour, the signals received by the eye are classified into red, green, and blue, denoted by the name tristimulus values. The tristimulus values of any particular colour are the three components of R, G, and B. The sensor that we use also provides signals similar to the tristimulus values.

From the values of R, G, and B received as Vr, Vg and Vb, one has to calculate two values known as 'chromaticity coordinates.' The formula for the calculation of these two x and y coordinates was introduced by the International Committee on Illumination or CIE in 1931. This committee, after taking the readings of thousands of visual observers with varied colour objects, arrived at a chart known as the Chromaticity chart, shown in Fig. 1. From the calculated x and y coordinates, the colour can be determined by plotting this (x, y) point in the chart.

$$x = Vr / (Vr + Vg + Vb)$$

$$y = Vg / (Vr + Vg + Vb)$$

For instance, x = 0.2 and y = 0.35 gives bluish-green colour and x = 0.47 and y = 0.45 gives yellow colour. The chart has a horseshoe shape, which is peculiar but is the result of an exhaustive study by CIE. The edge of the chart shows the purest of colours with the spectral wavelength values shown on certain points noted therein.

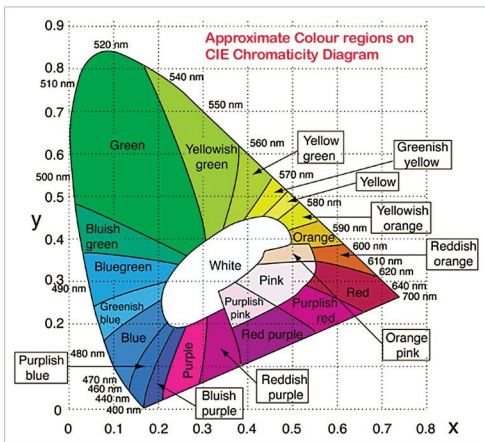


Fig. 1: Chromaticity chart with x and y being the red and green coordinates

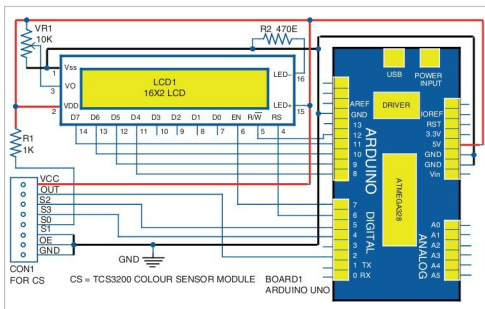


Fig. 2: Circuit diagram for colour identification